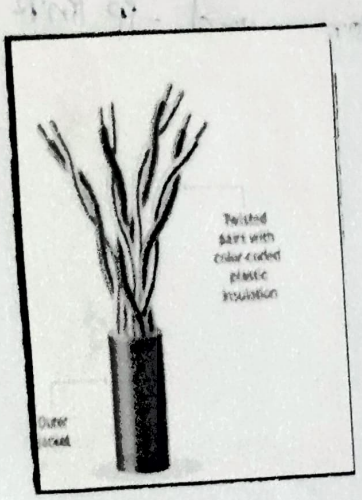
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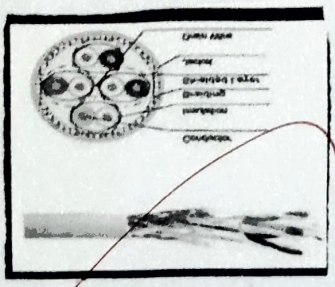
Study of different types of Network Cables

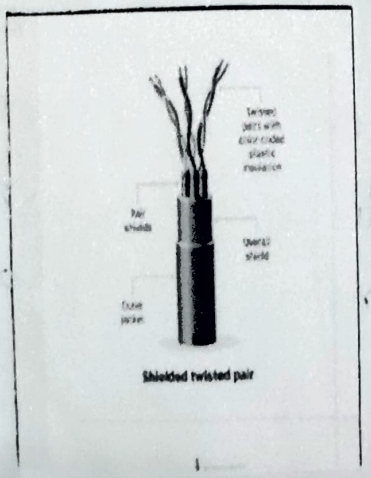
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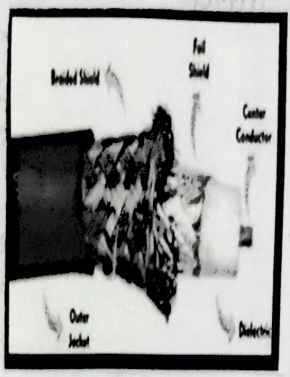
To understand different types of network cable.

Cable type	Category	Max data Transmission	Advantages/ Disadvantages	Application /use
UTP	Category3	10 bps	Advantages: * cheaper in cost * Easy to install as they have a smaller overall diameter. Disadvantages: * more prone to (EMI) Ethernet interference and noise	10 BaSe Ethernet
	Category5	100 mbps		Fast ethernet, Gigabit ethernet
	Category5e	1 GPS		Fast ethernet, Gigabit Ethernet
STP	Category 6, 6a	10 Gbps	Advantages: * Shielded * faster than UTP * Less susceptible to noise and interference. Disadvantages: * Expensive * Great for installation effort	Gigabit Ethernet, 10G Ethernet (55m) Widely used in data centres.
SSTP	Category 7	10 Gbps		Gigabit Ethernet 10 G Ethernet (100 m)
Coaxial Cable	RG-6 RG-59 RG-11		Advantages: * High bandwidth * Interface * Low loss * versatile Disadvantages: * Limited distance * Cost * Size is bulkey	Speed of Signal is 500m. Television network. High speed Internet connections.


Twisted pairs with color-coded plastic insulation
Outer shield




Twisted pairs with color-coded plastic insulation
Pair shields
Overall shield
Drain wire
Shielded twisted pair


Braided Shield
Foil Shield
Center Conductor
Outer Jacket
Dielectric



Cable Type	Category	Maximum data transmission	Advantages / Disadvantages	Applications / uses
Fiber optics cable	Single mode Multi mode	100 Mbps	Advantages: - High speed - High bandwidth - High security - Long distance Disadvantages: - Expensive - Required skilled installer	Maximum distance of fiber optical cable is around room.

Result:

Thus, we have learned the different types of network cable for computer networks.